

Major/Year_____ Student ID# _____ Name _____

Q1. 監督學習和無監督學習之間最明顯的區別是什麼？ (5%)

What is the most obvious distinction between the “Supervised Learning” and “Unsupervised Learning”

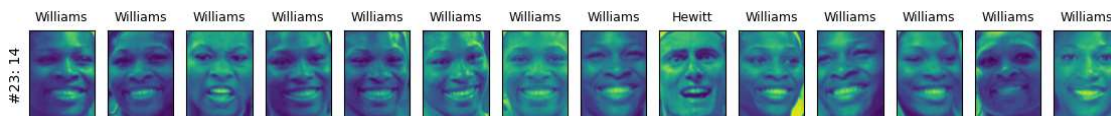
Q2. [] “email 作為輸入,判斷 email 是否是垃圾郵件” 屬於哪種學習？ (5%)

(1) 監督式學習 (Supervised) (2) 無監督式學習 (Unsupervised) (3) 都是 (4) 都不是

“an email as the input, judge if it is a spam email or not” is in the category of

(2) Supervised Learning (2) Unsupervised Learning (3) Both (4) Neither

Q3. [] “輸入 3023 張人臉照片,機器學習判斷以下照片屬於同一個人” 屬於哪種學習？ (5%)



(1) 監督式學習 (Supervised) (2) 無監督式學習 (Unsupervised) (3) 都是 (4) 都不是

“given 3023 photos of human faces, M.L. said these photos belong to one person”

is in the category of (1) Supervised Learning (2) Unsupervised Learning (3) Both (4) Neither

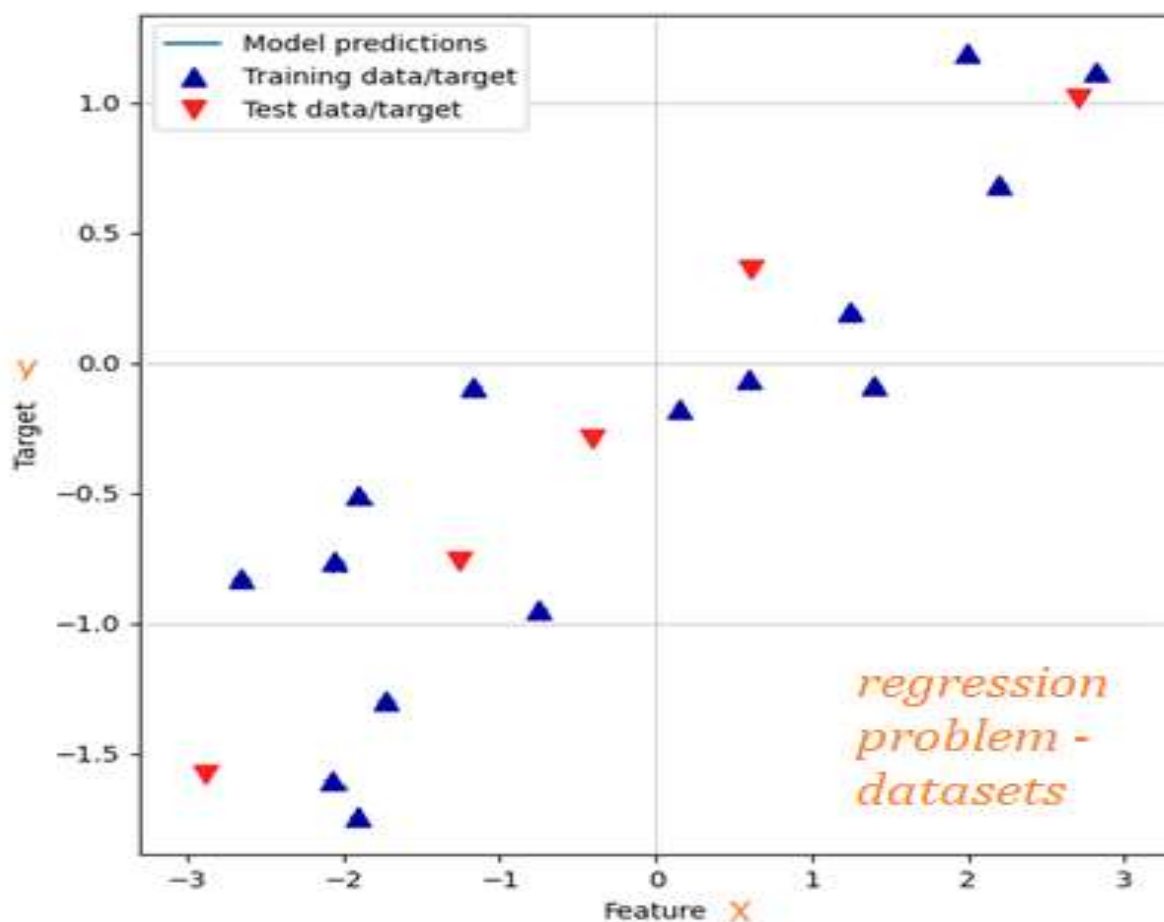
Q4. 我們將收集到的數據分為訓練數據集和測試數據集。We divide the collected data into training datasets and test datasets.

(a) 請解釋 “overfitting” 的含義 (Please explain the meaning of *overfitting*) (5%)

(b) 請解釋 “underfitting” 的含義 (Please explain the meaning of *underfitting*) (5%)

(c) 為什麼 “overfitting” 不好 (why is *overfitting* not desired) (5%)

Q5. Given training data (blue triangle) below, draw the **decision boundary line** (i.e. model prediction line) for this *regression* model if we use kNN method with only 1 nearest neighbor. 給定下面的訓練數據 (藍色三角形), 如果我們用 $kNN = 1$ 方法, 繪製回歸模型的**決策邊界線** (即模型預測線)。(5%)



Q6. Using **KNN** (K-Nearest Neighbors) algorithm with **60** pairs of data for a **regression** model, you are to read the data from the file “**wave60_dataset.txt**”, split them into the test datasets (20%) and training datasets (80%), and find the test dataset and training dataset scores with the following KNN parameters.

使用 KNN (K-Nearest Neighbors) 算法和 60 組數據作**回歸**模型，您將從 “wave60_dataset.txt”中讀取數據，將它們分為測試數據集（20%）和訓練數據集（80%），並使用以下 KNN 參數找到測試數據集和訓練數據集的分數。（65%）

n_neighbors = 1, 3, 5, 7, 9
weights = ‘uniform’ ‘distance’

In the *Python Shell Window*, print the test and training dataset scores in the way as shown below.

在 *Python Shell Window* 中，以如下的方式印測試數據集和訓練數據集的分數。

```
===== RESTART: C:/Python38/ml/nycu_midterm.py =====
uniform , KNN=1, X_test/X_train score = 0.36/1.00
uniform , KNN=3, X_test/X_train score = 0.72/0.85
uniform , KNN=5, X_test/X_train score = 0.67/0.83
uniform , KNN=7, X_test/X_train score = 0.66/0.79
uniform , KNN=9, X_test/X_train score = 0.65/0.77

distance, KNN=1, X_test/X_train score = 0.36/1.00
distance, KNN=3, X_test/X_train score = 0.55/1.00
distance, KNN=5, X_test/X_train score = 0.56/1.00
distance, KNN=7, X_test/X_train score = 0.61/1.00
distance, KNN=9, X_test/X_train score = 0.60/1.00
```

I have assumed that students are able to read text data file, therefore a Python program “**midterm.py**” is provided with the reading function in the E3 platform. You may use it to save time. Just a reminder – for the regression model the KNN module to use is “**KNeighborsRegressor**”. You can use “import” package to save time.

我假設學生能讀取數據，因此在 E3 平台中提供了一個具有讀取功能 Python 程序 “midterm.py”。您可以使用它來節省時間。提醒一下：對於回歸模型，KNN 模塊使用的是 “**KNeighborsRegressor**”。您可以使用“導入”包來節省時間。

(65% - reading 10%, using at least 1 loop (for or while) 15 %, kNN 15%, weights 15%, printing format 10%)